

Paper Text

Introduction

A. Motivation and aims

- Tokamak - importance of poloidal coils optimization
- Different objectives to minimize
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- Aims of the paper:
 - Compare Bayesian to LBFGS
 - Create a Bayesian algorithm that works for different machines (DIIID, MANTA etc) and cost functions, including fixed-boundary and free-boundary solves, producing more optimal coil configurations than other more computationally intense optimization schemes
- Want this to be used for new tokamak designs

B. Related work

Rail based methods (to reduce degrees of freedom)

- Reliance on estimated objective function gradients
 - High number of function calls necessary, even if estimated quasi-newton
- gradient-free global optimization, can handle diverse constraints, and uses a surrogate model of the multi-output objective function to intelligently guide function evaluations. Exploitation, exploration (read more papers on this)

BO:

- Different related work (relevancy!) 1
- Same but stellarator
 - <https://arxiv.org/pdf/2110.07464>
- Emphasize difference to https://scientific-publications.ukaea.uk/wp-content/uploads/072507_1_5.0272085.pdf
 - Getting information from fixed to inform our free solve without the need of a classifier
-

The work is inspired by Jeremy Salm's:

file:///Users/alt2005/Downloads/Submission_to_AIP_Salm%20(1).pdf

Related Work

Prior work on PF coil configurations spans several methodological families, each with limitations our approach attempts to address. The first type of methods attempts to reduce the search space. The very first methods fixed the placement of the coils and optimized over the current used [cite]. Furthermore, to allow coils to move, rail-based parametrizations, used in integrated design frameworks such as BLUEPRINT/Bluemira [cite Coleman & McIntosh 2020; Nilima 2025] and FUSE [cite Meneghini 2024], constrain coils to lie on a one-dimensional contour around the plasma. This process reduces dimensionality and simplifies engineering-constraint enforcement through exclusion zones. However, optimal configurations off the rail are unreachable by construction, and the gradient-based optimizers typically paired with these parametrizations can struggle on non-convex landscapes with competing objectives.

Other methods remove this geometric restriction. For instance, An et al. [cite An 2006] on EAST uses a genetic-based algorithm and the recent SCOPE framework [cite SCOPE 2025] uses simulated annealing, which do not need any assumptions on geometry but both require many function evaluations to converge. This issue becomes prohibitive with more difficult objectives, when one function evaluation becomes more computationally expensive. Large-scale Monte Carlo sampling with heuristic post-selection, as in the Fiesta-based STEP studies of Hudoba et al. [cite Hudoba 2023, 2024], face similar scaling challenges.

This work is directly inspired by Salm et al. [cite Salm 2026], who formulated PF placement as a two-level problem in which an inner regularized least-squares solve analytically determines optimal currents for a fixed geometry, while an outer L-BFGS-B loop [cite Byrd 1995] adjusts the geometry to minimize a fixed-boundary flux residual. Applied to the MANTA negative-triangularity pilot plant [cite MANTA 2024], the method achieved a 37% reduction in total PF current relative to the baseline. We retain this two-level structure but address two limitations. First, as it is a gradient-descent method, multi-start L-BFGS-B is local and sensitive to initialization: achieving global coverage requires many restarts, and the method provides no principled balance between exploration and exploitation. Second, though Salm et al. did validate the final coil configurations with a free-boundary Grad-Shafranov solve, the solution finds an optimum in the fixed-boundary cost function rather than the free-boundary cost function, the former of which is only a proxy for the latter. The issue is that a geometry that matches target flux values at discrete boundary points in the fixed boundary case can still produce a distorted or shifted LCFS when the coupled plasma-coil physics is solved. We address the first by replacing the outer loop with Bayesian optimization, and the second by extending the cost to blend the computationally cheap fixed-boundary residual with an expensive free-boundary LCFS mismatch term.

BO has been successfully applied across fusion design and control, establishing it as a viable methodology for expensive simulation-driven problems. Applications include rampdown design on DIII-D [cite Mehta 2024], disruption mitigation on ITER [cite Pusztai 2023], runaway-electron calibration [cite Järvinen 2022], divertor monoblock stress [cite Humphrey 2024], and current-drive profile design for STEP [cite Brown 2024]. In stellarator coil design, BO has been applied in several works [cite Giuliani 2024; Glas 2022], including the two-stage architecture of Glas et al., which performs global BO followed by local gradient-based refinement — an approach we adopt here. Like Glas et al., we use gradient-free global optimization with a GP surrogate to guide evaluations, then refine promising candidates with local gradient-based optimization. **Write about how your paper is different**

The closest methodological work is Nunn et al. [cite Nunn 2025], who apply multi-objective BO to the STEP PF coil system using a free-boundary solve. Because free-boundary solves frequently fail in portions of the input space, they train a separate Gaussian process classifier to predict failures and mask the acquisition function accordingly — an approach also used **by Hornsby et al. [cite Hornsby 2024] for gyrokinetic datasets**. This additional complexity entails a new model that must itself be trained, validated, and tuned. The classifier’s precision was 82% and recall was 86%, meaning feasible regions are sometimes incorrectly excluded. Our combined-boundary formulation sidesteps the classifier entirely. By informing the free-boundary solve by blending its cost with the **better behaved** fixed-boundary residual while also considering that configurations that cause free-boundary failure typically also have poor fixed-boundary scores, BO can steer away from failure regions without an auxiliary model. Beyond this methodological difference, we also stress-test across a broader regime than Nunn et al., who fix their setup to the STEP baseline with seven coils confined to pre-specified permissible zones: we do not confine coils to narrow regions, and we sweep over the number of coils and the regularization weights for current in both the fixed and free-boundary solve, stress testing the optimizer and characterizing the solution’s dependence on each hyperparameter.

Outline

Background

A. Axisymmetric ideal MHD equilibria

B. TokaMaker

Optimization Problem

A. Search Space and Parametrization

Before defining the optimization procedure, we first specify the admissible region in which PF coils are allowed to move. We define the coil position space as a continuous region bounded by two user-defined curves in the poloidal cross-section: an inner boundary (typically close to the limiter or vessel) and an outer boundary reflecting engineering constraints. This construction allows flexible geometric constraints while maintaining a compact search domain.

To reduce dimensionality, we assume up–down symmetry of the equilibrium. As a result, only coils in the upper plane need to be parameterized, with the lower coils determined by reflection. This reduces the number of free parameters by a factor of two. This methodology and the provided code not only extend to asymmetric configurations but allow for the user to define spaces to constraint specific coils.

Each coil location is parameterized by a $\theta \sim = (\theta, \mu) \in [0, 180]^\circ \times [0, 1]$. θ is the poloidal angle in degrees, defining position along the contour, and μ is a radial parameter which is unitless being invariant to size of reactor. The coil's θ being above 90° and below 90° correspond to the coil being to the left and to the right of the Z-axis respectively. The μ is equal to 1 at the outer bounding curve and 0 at the inner bounding curve. Thus, the position of a coil is given by a mapping from this normalized coordinate system into physical space. For a configuration of $2N$ coils total, the full parameter space is:

$$\Theta = (\theta_1, \dots, \theta_N, \mu_1, \dots, \mu_N)$$

This parameterization defines a compact, bounded search space amenable to both gradient-based and global optimization methods. **Mirroring stuff**

Finally, the coil position space is discretized for numerical evaluation in TokaMaker. Too coarse a grid may obscure local minima, while excessively fine resolution increases computational cost. Thus, the resolution is treated as a hyperparameter.

B. Optimization Objective

The main objective is to identify coil configurations that reproduce a target plasma equilibrium while minimizing coil currents and ensuring that solutions are physically feasible. With that in mind, this method was chosen to be the following: it is a bilevel optimization, in which the inner loop solves a Least Squares by computing the sets of currents per coil that would best reconstruct the desired Last Closed Magnetic Flux Surface (LCFS) while regularizing via L2 to keep the sum of currents low. The outer loop entails finding coil coordinates for a given set of currents that best reconstruct the LCFS while penalizing proximity of coils. The latter is implemented as a physics-informed heuristic: well-distributed coils tend to produce desired magnetic configurations. [cite]

Defining Psi

Defining Green Matrix

Thus, for a given set θ , the inner loop cost function is defined as:

- Inner cost
- Optimal current
- Fixed-Boundary Cost
- Free-Boundary Cost
- Coil Separation Penalty
- Normalization
- Final Objective: $\Theta = \arg \min J(\Theta)$

B. Bayesian Optimization

Three stages

- I. Bayesian (flow diagram) Loop
 - A. Initial sample (pre phase 1)
 1. Sobol sampling
 2. Symmetry of coils ($n_{\text{initial}} * n_{\text{coils}}!$)
 - B. Phase 1:
 1. Fit Gaussian Process
 2. Maximize EI acquisition function
 3. Call actual cost function at that optimum
 4. Stop phase 1 when improvement in the running best value falls below the convergence threshold over a fixed stagnation window.
 - C. Phase 2:
 1. Deduplication in physical R,Z space
 - a) Keep unique candidates to inform local refinement
 2. Local refinement via LBFGS

D. Remarks

1. The surrogate models the full combined objective, not just fixed-boundary error.
2. BO is especially useful here because the objective is:
 - a) expensive,
 - b) nonconvex,
 - c) partially nonsmooth,
 - d) and may incur free-boundary failures.
3. The method is gradient-free in the global phase, so it does not require differentiating through the free-boundary solve.
4. The local L-BFGS-B phase is used only after BO has already identified promising regions.

DIID Case Study

A. Visuals of how bayesian works

- Cost landscape visualization to show BO's surrogate model evolving over iterations on the 2D DIIID landscape. This motivates why BO works here (the cost is smooth enough for a GP)
 - Smoothness functions
 - 1D plot, where you slice and then vary the r and θ

B. Comparison of Sobol Sampling, BO, and LBFGS

Fixed

- Compare to brute force
- LBFGS
 - Broyden–Fletcher–Goldfarb–Shanno algorithm (BFGS), a quasi-Newton method accessible in the Python library SciPy. LBFGS
- Less function calls, similar cost
 - Performance under different coils, different λ s
 - Better under certain configurations?
- Time for both
 - Consider if function calls took longer to call? N seconds per function
 - As $n \rightarrow \infty$, Bayesian overhead ($O(\text{points in it}^3)$) is negligible

C. Hyperparameter Selection

- n_{initial} choice
- Symmetry trick excluded
- Influence of the regularization parameter λ
- Influence of acquisition function EI, PI, LCB

Conclusion

References

https://scientific-publications.ukaea.uk/wp-content/uploads/072507_1_5.0272085.pdf

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